Materials & Design 245 (2024) 113265

Contents lists available at ScienceDirect

Materials & Design

journal homepage: www.elsevier.com/locate/matdes

Reduced warpage in semiconductor packages: Optimizing post-cure temperature profile considering cure shrinkage and viscoelasticity of epoxy molding compound

Hui-Jin Um [a], Young-Min Ju [a], Dae-Woong Lee [b], Jinho Ahn [c,d,*], Hak-Sung Kim [a,d,e,*]

[a] Department of Mechanical Engineering, Hanyang University, 222, Seongdong-gu, Seoul 04763, Republic of Korea
[b] SK Hynix Inc. Icheon, Gyeonggido, Republic of Korea
[c] Division of Materials Science and Engineering, Hanyang University, Seoul 04763, Republic of Korea
[d] Hanyang Institute of Smart Semiconductor, Hanyang University, 222, Wangsimni-ro, Seongdong-gu, Seoul, Republic of Korea
[e] Hanyang Research Center for Advanced Semiconductor Packaging, Hanyang University, Seoul 04763, Republic of Korea

ARTICLE INFO

Keywords:
Epoxy molding compound
Cure shrinkage
Warpage
Finite element analysis
Optimization

ABSTRACT

Warpage in semiconductor packages is a critical issue that affects their reliability and performance. This study aims to minimize the warpage of a bi-material dummy package by optimizing the post-mold curing (PMC) temperature profile. A warpage simulation model was developed considering the viscoelastic properties and cure shrinkage behavior of the epoxy molding compound (EMC). A compression molding monitoring system was also developed to analyze the cure behavior of the EMC. The gelation point of the EMC during compression molding was determined using a dielectric sensor, while cure shrinkage behaviors were analyzed using a fiber Bragg grating sensor. Viscoelastic properties were characterized through stress relaxation tests, and thermal properties were measured using three-dimensional digital image correlation. The developed simulation model was used to optimize the PMC temperature profile using a genetic algorithm. Experimental validation showed that the optimized temperature profile reduced the warpage from ~1519 μm to 486 μm compared to a linear profile. The findings of this study have broader implications for the design and manufacturing of semiconductor packages, as the proposed approach can be applied to minimize warpage and improve package reliability.

1. Introduction

The advancement of semiconductor packaging technology represents a crucial area in the semiconductor industry. The management of warpage in package structures has become a growing concern with this technological advancement [1–3]. Package structures are composed of diverse materials and are susceptible to warpage due to differences in material properties during temperature changes in the manufacturing processes. Recently, wafer-level packaging has received much attention. However, in wafer-level packaging, excessive warpage can occur due to the large wafer size and the complicated fabrication process [2,4–7], which can induce defects such as interfacial delamination and can cause significant reliability challenges for package structures. Therefore, accurate prediction and effective control of warpage are essential for successful manufacturing of a semiconductor package.

In semiconductor packaging, one of the many factors that can cause warpage is cure shrinkage of the epoxy molding compound (EMC). The EMC is composed of a variety of materials, such as epoxy resin, filler (silica), and additives [8,9]. In the EMC, notable shrinkage arises from the chemical and thermo-mechanical interactions involved in molecular cross-linking during the curing and post-mold curing (PMC) processes. Considering that the mechanical elastic modulus develops following the gelation, it is necessary to analyze the effective cure shrinkage after gel point [10–13]. Therefore, the cure shrinkage of EMC that occurs during compression molding and the PMC process should be evaluated to predict the warpage of the actual package structure. It is also important to consider the viscoelastic properties of the EMC, in which stress relaxation occurs with time and temperature. If the viscoelastic properties are not considered, warpage can be inaccurately predicted, and even its direction, either negative (smiling warpage) or positive (crying

* Corresponding authors at: Division of Materials Science and Engineering, Hanyang University, Seoul 04763, Republic of Korea (J.Ahn) and Department of Mechanical Engineering, Hanyang University, 222, Seongdong-gu, Seoul 04763, Republic of Korea (H.S. Kim).
E-mail addresses: jhahn@hanyang.ac.kr (J. Ahn), kima@hanyang.ac.kr (H.-S. Kim).

https://doi.org/10.1016/j.matdes.2024.113265
Received 17 June 2024; Received in revised form 1 August 2024; Accepted 19 August 2024
Available online 22 August 2024

warpage), may be predicted incorrectly [14–17].

Huang et al. investigated the cure shrinkage behavior of EMC during the initial molding and subsequent PMC processes using differential scanning calorimetry and thermal mechanical analysis [18]. Their findings highlight the significance of considering the non-negligible strain (∼0.01–0.035 %) during the PMC process when designing the packaging process. In a similar context, Hasegawa et al. demonstrated the importance of filler distribution optimization in EMC for reducing warpage and enhancing reliability in fan-out wafer-level packaging [19], highlighting the critical role of material design in achieving desired package performance. Numerous studies have employed finite element (FE) simulations to analyze various characteristics of EMC, such as cure shrinkage, viscoelastic properties, and warpage [10–17,20–22]. These simulations offer valuable insights into the complex behavior of EMC and provide guidance for the design of packaging materials and structures. Building upon these findings, researchers have explored strategies to reduce warpage by modifying materials, such as fillers or resins [19,23,24], or adjusting structural dimensions, including thickness and material proportions within the package [6,7,25]. While these studies have significantly contributed to the understanding of EMC cure shrinkage and warpage behavior, the primary focus has been on material modifications and package structure optimization. Although some studies have investigated the influence of process parameters on warpage, the results consistently indicate that increasing PMC time leads to reduced warpage [26]. This suggests a limitation in process time, as longer PMC durations are required to minimize warpage effectively.

In this study, the temperature profile of the PMC process was optimized to minimize the warpage in a package structure manufactured with compression molding and PMC processes. A compression molding monitoring system was devised in which EMC cure shrinkage were monitored using a fiber Bragg grating (FBG) sensor, and the gelation point of EMC was detected using a dielectric sensor. Additionally, the stress relaxation determined from three-point bending tests at different temperatures was used to analyze the viscoelastic behavior of the EMC. A warpage prediction model for bi-material dummy packages was developed using the obtained EMC properties, and the model was validated by measuring the warpage of a bi-material dummy package using a three-dimensional (3D) digital image correlation (DIC) technique.

2. Material characterization

2.1. Evaluation of the effective cure shrinkage of the EMC

A granular type of the EMC was used to analyze its mechanical behavior during the compression molding process. As shown in Fig. 1(a), the granular EMC had an irregular cylindrical shape with an average length and diameter of 3.36 and 1.17 mm, respectively. These EMCs were primarily composed of silica and epoxy, and the composition ratio of each material was analyzed with images from a scanning electron microscope (Quattro ESEM, Thermo Fisher Scientific, USA), as presented in Fig. 1(b). Spherical silica, shown in bright colors, was observed, and the size of the silica ranged from nm to µm. The composition ratio of the silica was 84.67 %.

The EMC specimens were fabricated with a two-step process, as shown in Fig. 1(c). During the curing step, compression molding was performed to fabricate the specimens, applying 7 MPa pressure at 165 °C. After curing, the specimen was cooled to room temperature and the PMC was performed for 2 h at 180 °C to achieve complete curing. A compression molding monitoring system was devised to monitor the status of the EMC during curing and comprised four parts as shown in Fig. 2(a): the EMC mold, temperature control, pressure control, and data monitoring. Fig. 2(b) shows the cylindrical EMC specimen fabricated using a compression molding monitoring system. The metal mold for the EMC specimen was composed of several parts to prevent leakage of the EMC under high temperature and pressure. Gaskets and rubber rings

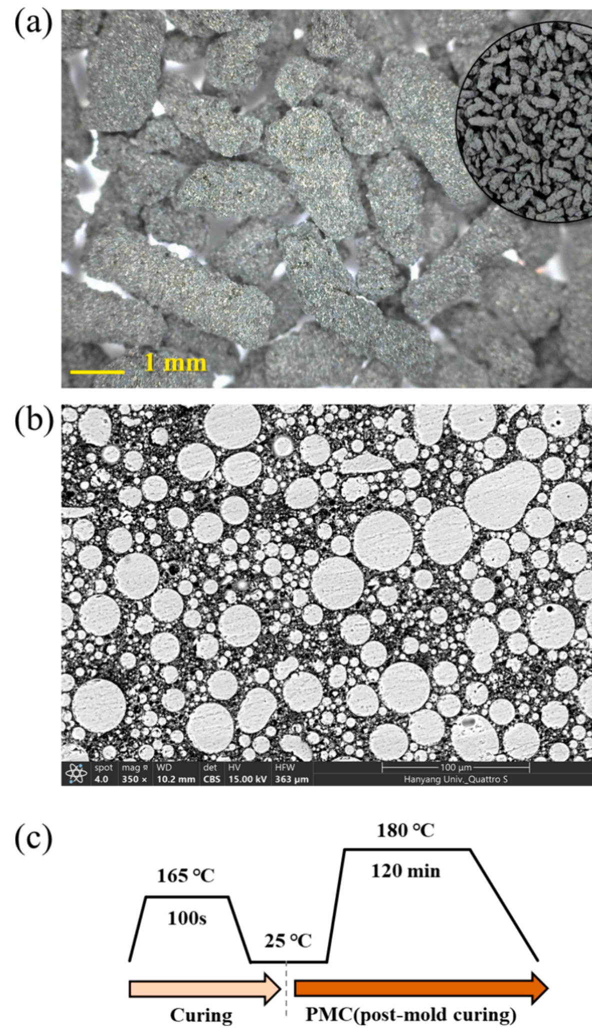

Fig 1. (a) Granular type epoxy molding compound (EMC) and (b) SEM image analysis in cross-section view of EMC. (c) Schematic of EMC cure processes.

were used at the bottom and top of the mold, respectively. The granular EMC was inserted into a cylindrical metal mold that was placed into a ceramic thermal chamber and heated to the target temperature (165 °C) and a simultaneous hydrostatic pressure of 7 MPa. Three types of sensors were utilized to monitor the EMC properties. In situ temperature monitoring was conducted with a K-type thermocouple and a DAQ 9211 with LabVIEW system. The dielectric sensor was used with an LCR meter (U1732C, Agilent Technologies, USA) to determine the gel point of the EMC. Because cure shrinkage occurred after the gel point, determination of the gel point of the EMC was essential [27]. By applying an alternating current at 1000 Hz to the EMC during the curing process, variations in ion viscosity (resistivity) could be measured in real time, allowing determination of the gel point of the EMC. The dielectric sensor consisted of parallel plate electrodes with intersecting Cu patterns. The dissipation factor (D) representing the mobility of ions within the EMC was measured using the dielectric sensor, as expressed by the following equation.

$$D = \left|\frac{I_R \bullet V_m}{I_C \bullet V_m}\right| = \left|\frac{I_R}{I_C}\right| = \left|\frac{Z_C}{Z_R}\right| = \frac{1}{\omega R_m C_m} \quad (1)$$

where I stands for the electric current, Z represents the equivalent impedance, V_m denotes the alternating voltage applied to the equivalent circuit with angular frequency ω, and the subscripts R and C correspond to the resistance and capacitance of the equivalent circuit model, respectively. The gelation point was defined at a derivative value of zero

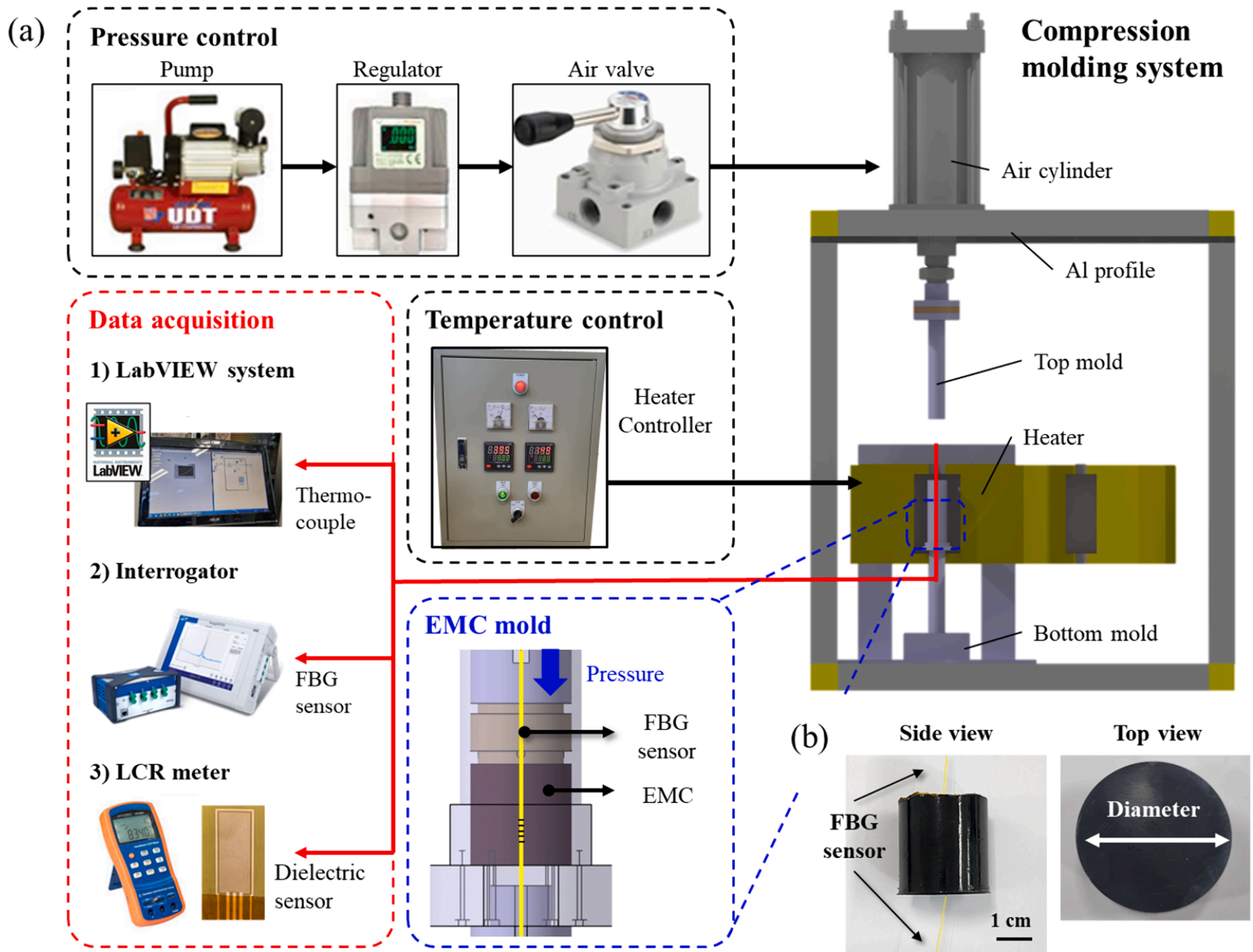

Fig. 2. (a) Schematic image of compression molding system for cure of epoxy molding compound (EMC). (b) Fabricated EMC specimen.

[28]. After gelation, cure shrinkage of the EMC occurred and was measured with the FBG sensor. FBG sensors have a Bragg pattern that depends on temperature or mechanical load, allowing strain to be measured. The Bragg wavelength (λ_B) of an FBG sensor was described using the relationship between the effective refractive index (n_{eff}) of the fiber core and the grating pitch (Λ) of fiber.

$$\lambda_B = 2n_{eff}\Lambda \tag{2}$$

The shift in the Bragg wavelength ($\Delta\lambda_B$) caused by external disturbances such as stress or temperature can be expressed as

$$\Delta\lambda_B = \Delta\lambda_B^i + \Delta\lambda_B^d \tag{3}$$

where $\Delta\lambda_B^i$ and $\Delta\lambda_B^d$ are intrinsic λ_B shifts related to deformation caused by temperature change or mechanical stress, respectively. Here, $\Delta\lambda_B$ is a value obtained through experimentation. If the temperature is measured during the experiment, $\Delta\lambda_B^i$ can be derived using the correlation equation between temperature and λ_B. Therefore, using the above relationship, $\Delta\lambda_B^d$ can be obtained by subtracting $\Delta\lambda_B^i$ from $\Delta\lambda_B$. When the ratio of the diameters of the EMC and the FBG sensor (β) is greater than 200, it can be assumed that only uniaxial strain occurs along the longitudinal direction of the fiber [10,11]. Therefore, the strain of the EMC could be expressed using deformation λ_B shifts as

$$\varepsilon = \Delta\lambda_B/\lambda_B(1 - P_k) \tag{4}$$

$$P_k = -\frac{n^2}{2}\{P_{12} - \upsilon_f(P_{11} + P_{12})\} \tag{5}$$

where P_k is the equivalent strain-optic constant, which consists of the effective refractive index of the fiber core and cladding materials (n), the strain optic constants of the Bragg grating (P_{11} and P_{12}), and the Poisson's ratio of the fiber (υ_f). P_k is a unique optical property of the FBG sensor, and a reference value of 0.202 was used in this experiment [29].

To sufficiently reflect the EMC strain in the FBG sensor, the specimen dimensions were determined considering the difference in stiffness between the FBG sensor and the EMC. An FE model of an EMC specimen was used, as shown in Supplementary Fig. S1(a and b). A schematic of the FBG sensor with which the strain within the Bragg grating pattern was analyzed is presented in Fig. 3(a). The diameter of the EMC specimen was incrementally increased while maintaining the FBG sensor diameter, and the specimen was subjected to a temperature gradient of 50 °C. The mechanical properties used in the FE analysis are shown in Table S1. Fig. 3(b) presents the results of the FE analysis, showing the EMC thermal strain detection by the FBG sensor as a function of EMC specimen diameter. A specimen diameter of 25 mm was selected to ensure strain measurement accuracy within a 1 % error range. This diameter minimizes the strain difference between EMC modulus of 1 GPa and 5 GPa to 10^{-5} %, which is not achieved with smaller diameters. Prior to measuring the cure shrinkage with the FBG sensor, the correlation between temperature (T) and λ_B^i of the FBG sensor was analyzed

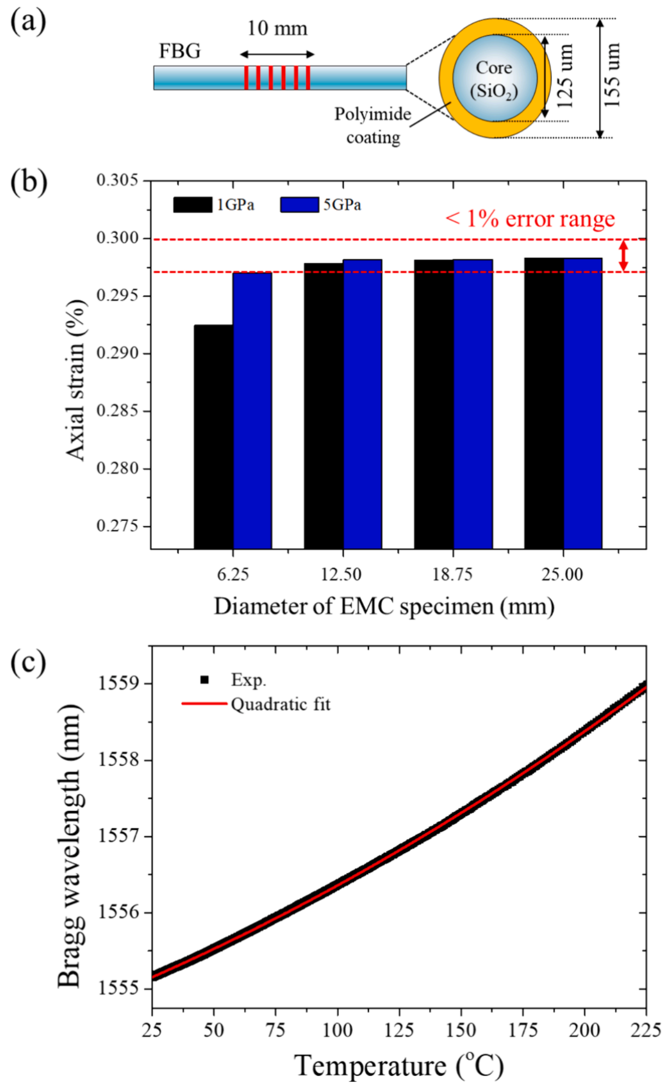

Fig 3. (a) Schematic of FBG sensor structure with Bragg grating pattern (length of 10 mm). (b) Axial strain according to the elastic modulus and diameter of EMC specimen. (c) The relationship between the Bragg wavelength and temperature.

[30–34], as shown in Fig. 3(c). The dependence of λ_B^i on T (in °C) could be expressed as

$$\lambda_B^i = 1554.8155 + 0.0131T + 2.3869 \bullet 10^{-5}T^2 \quad (6)$$

In the experiment, only EMC cure shrinkage was analyzed, as the effect of temperature could be considered by the above relationship (λ_B^i). After the compression molding monitoring experiments, the specimens were removed from the mold and the PMC was performed in a thermal chamber. The real-time temperature and EMC cure shrinkage were also measured during PMC process.

2.2. Viscoelastic properties of the EMC

A stress relaxation experiment was performed to analyze the viscoelastic properties of EMC as shown in Fig. 4. The EMC specimen was fabricated into a thin plate shape following the two-step manufacturing process of curing step and post-mold curing as specified in section 2.1. The dimensions of the thin, plate-shaped EMC specimens were 35 × 45 × 0.55 mm^3. The stress relaxation tests were performed using a micro-tensile machine (Lloyd Instruments, USA). The extracted

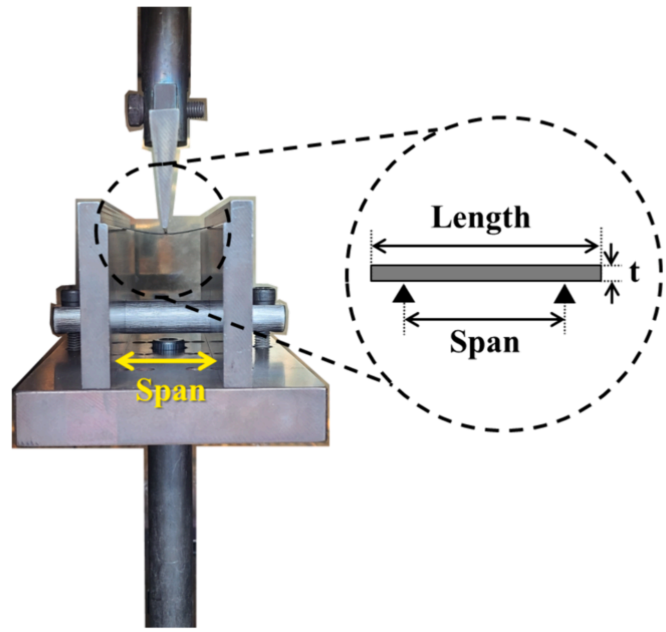

Fig. 4. Stress relaxation test setup and schematic of specimen.

load–displacement (LD) curve generated from stress relaxation test was used to determine flexural modulus (E_f) with the following equation [35].

$$E_f = L^3m/4bd^3 \quad (7)$$

where L represents the span length, m is the initial linear region gradient on the LD curve, b is the specimen width, and d is the specimen thickness. Stress relaxation experiments were conducted at a span-to-depth ratio of 40. The tests were performed by applying a displacement load at the midspan to achieve a strain of 0.5 % while varying the temperature from room temperature to 260 °C. At each temperature, a dwelling time (30 min) was maintained to ensure stabilization before measuring stress relaxation behavior.

2.3. Thermal characteristics of the EMC

The thermal expansion coefficient of the EMC is different at temperatures below and above the glass transition temperature (T_g). Therefore, the thermal expansion coefficient was measured with the EMC specimen with a size of 40 × 40 mm^2 and a thickness of ~0.55 mm. The specimen was placed in a thermal chamber and the thermal expansion coefficient of the EMC was measured using the 3D-DIC (ARAMIS Adjustable, ZEISS, Germany) technique. The measurements were conducted by gradually heating the specimen from room temperature to 200 °C to cover the entire temperature range of the curing process. To validate the CTE measurements obtained through DIC, a complementary analysis using thermomechanical analysis (TMA) was conducted on the EMC specimens. TMA measurements were performed over a temperature range from room temperature to 250 °C, with a heating rate of 2 °C/min.

3. Material properties and behavior

3.1. Cure shrinkage of EMC during compression molding

Fig. 5(a) shows EMC specimens fabricated with the compression molding and PMC processes in a cylindrical shape with a diameter of 25 mm. Based on an optical microscope analysis of the central cross section, the FBG sensor was located in the center of the specimen. Prior to the analysis of effective cure shrinkage, the gelation point of the EMC was

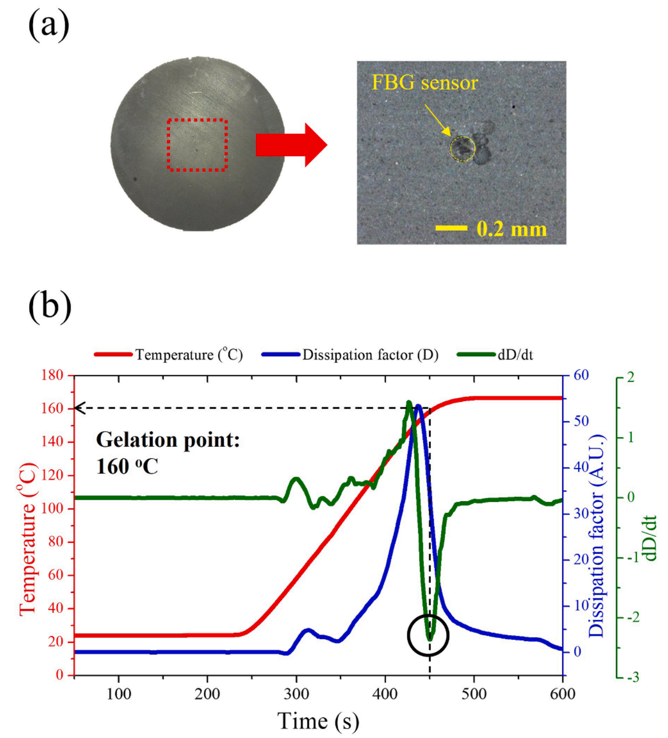

Fig. 5. (a) Optical image of cross section view for EMC specimen. (b) Experimental results of EMC compression molding: temperature profile, dissipation factor (D), and rate of dissipation factor (dD/dt).

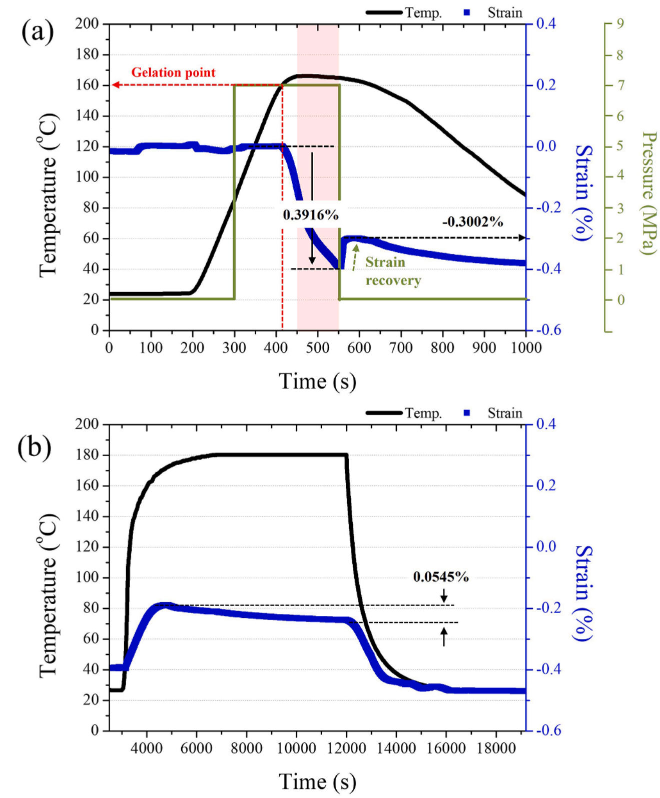

Fig. 6. Experimental results of cure shrinkage measurement for EMC: (a) Compression molding process, and (b) post-mold curing process.

defined based on the results of the dielectric sensor, as indicated in Fig. 5(b). While heating from room temperature, the dissipation factor increased and a maximum peak was observed, after which the rate of dissipation reached its minimum at ~160 °C. In general epoxy, the curing reaction initiates when dD/dt reaches its maximum and concludes when dD/dt reaches zero [36]. Thus, the reaction rate peaks when the resin viscosity is at its lowest, and the point where dD/dt intersects zero coincides with the resin's lowest viscosity due to the greatest movement of dipoles and ions induced by the supplied energy. Furthermore, gelation of the epoxy resin occurs when dD/dt reaches its minimum, at which point the degree of cure is ~0.6 regardless of the rate of temperature increase [28]. From the results, the temperature of 160 °C was defined as the gelation point at which effective cure shrinkage began to occur.

The effective cure shrinkage of the EMC occurring during the curing process is presented in Fig. 6. Even though pressure was applied starting when the temperature reached ~80 °C during compression molding, EMC strain began to occur after the gelation point predefined through dielectric sensor analysis. The consistent results between the dielectric and FBG sensors confirmed that strain occurred within the EMC only after the gelation point. After reaching the target curing temperature, the amount of cure shrinkage strain continued to increase even during the cure process when the temperature was maintained because cure shrinkage occurred as cross-links were formed during curing of the EMC [12,18]. It was noteworthy that the amount of strain decreased upon the removal of pressure, as indicated in Fig. 6(a), which meant that the elastic deformation induced by the external pressure was recovered. Therefore, the deformation amount excluding the corresponding elastic strain from the total strain could be regarded as the effective cure shrinkage. Consequently, the effective cure shrinkage of the EMC used was −0.3002 %. During the final cooling stage, thermal shrinkage strain occurred as the temperature decreased.

Once the compression molding cure process was completed, the EMC specimen was removed from the mold and the PMC process was performed in the thermal chamber. The strain data obtained during this stage are shown in Fig. 6(b). During both the heating and cooling stages, strain by thermal deformation of the EMC was observed. The amount of additional cure shrinkage strain (0.0545 %) was consistent even during the temperature holding stage in PMC process, as was reflected in the FE model.

3.2. *Viscoelastic behavior and thermal characteristics of the EMC*

The results of the stress relaxation tests at different temperatures are presented in Fig. 7(a). Stress relaxation occurred over time at all temperatures and became more pronounced near the T_g of EMC (145 °C). To account for this behavior, 145 °C was chosen as the reference temperature, and the stress relaxation data at each temperature were shifted along the time axis accordingly. A Prony series fitting was then performed, as shown in Fig. 7(b). The degree of shift at each temperature was expressed using the WLF equation, wherein it was evident that the shift factor behavior differed below and above the T_g (Fig. 7(c)). Therefore, to express the viscoelastic behavior more accurately, separate WLF fittings were performed below and above the reference temperature, and the ABAQUS user subroutine was used to reflect each WLF coefficient in the viscoelastic properties (see Table 1). The thermal strain measurements using DIC with respect to temperature are shown in Fig. 7(d). Around 145 °C, the DIC results revealed a change in the slope of the strain, indicating that the temperature corresponded to the T_g.

4. Warpage analysis and optimization

4.1. *Warpage measurement of a bi-material dummy package specimen*

The bi-material dummy package specimen was used to analyze warpage behavior induced by the EMC during the curing process. To generate clear warpage differences in the limited lab-scale experiments, bi-material dummy package specimens were fabricated using aluminum

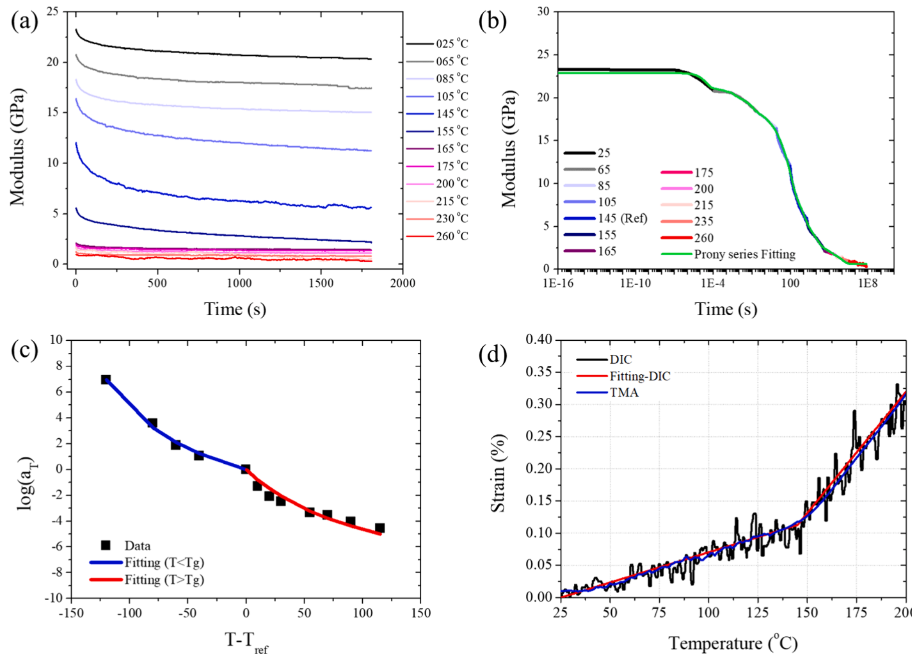

Fig. 7. (a) Stress relaxation test results according to the temperature. (b) Master curve as a result of stress relaxation test. (c) Shift factors of stress relaxation master curve and WLF fitting results before and after the glass transition temperature of EMC. (d) Thermal strain results using 3D-DIC.

Table 1
Viscoelastic properties of EMC as a results of stress relaxation test.

Prony series coefficients		
i	ρ_i	E_i **[GPa]**
∞	–	0.6007
1	3E-5	1.804
2	5E-4	0.101
3	5E-3	0.914
4	1E-1	1.5623
5	1E+0	0.8386
6	1E+1	2.1057
7	1E+2	6.4895
8	1E+3	3.7368
9	1E+4	2.5988
10	1E+5	0.7698
11	1E+6	1.3362
WLF coefficients	$T_{ref} = 145$ °C	
$T<T_{ref}$	C_1	C_2
	5.41	212.81
$T>T_{ref}$	C_1	C_2
	10.23	120.12

strips, as indicated in Fig. 8. The thickness of each material was selected to maximize warpage based on Timoshenko beam theory [37]. The UVC surface treatment (wavelength: 254 nm, 8 W, Hansung Ultraviolet Co.) was conducted for the aluminum strip to guarantee the wettability of the molten EMC before the compression molding process, as shown in Fig. 8 (a). After placing the UV-treated aluminum strip and the EMC in a metal mold, the specimen was treated with the same temperature and pressure conditions as in the compression molding process. As shown in Fig. 8(b), EMC case t1 was a specimen with an aluminum (0.20 mm)/EMC (0.16 mm) thickness ratio that maximized the warpage, and EMC case t2 with an aluminum (0.20 mm)/EMC(0.56 mm) thickness ratio was a comparative case used to validate the warpage based on the actual EMC thickness. After the compression molding process was completed, the warpage generated during the PMC process was measured with the 3D-DIC technique (see Fig. 8(c)).

4.2. Finite element analysis of the bi-material dummy package model

The bi-material dummy package structure consisting of EMC and Al was modeled to simulate the warpage behavior during compression molding and the PMC process, as shown in Fig. 9(a). As mentioned above, there were two cases based on the EMC thickness. The 3D solid element (C3D8R) was used and 32,400 elements (39494 nodes) were generated. The FE analysis was performed using ABAQUS (SIMULIA, USA) combined with a user temperature–time shift routine (UTRS) for time domain viscoelastic materials. As shown in the schematic of the overall FE analysis procedure in Fig. 9(b), there were two parts according of the cure process of the EMC. For the compression molding process, the cure behavior of the EMC was analyzed only after the gelation point of the EMC. Because the elastic modulus was almost zero before the gelation point, internal stress could be neglected in the EMC. Dielectric sensor measurements and DSC analysis (see Fig. S2) indicated that the EMC curing process was completed during the compression molding cycle. Consequently, for warpage modeling, an elastic modulus of 2.143 GPa, corresponding to the fully cured EMC at 165 °C, was assigned at the end of the compression molding process. The expansion constants were used to deal with the cure shrinkage of the EMC during the compression molding process. The bi-material dummy package structure was assigned X- and Y-symmetry conditions on its side surfaces and Z-symmetry conditions on its bottom surface to simulate the curing process during compression molding. A pressure load of 7 MPa was applied to the top surface. The pressure was maintained while the temperature was held constant at 165 °C, and then the pressure was removed once the cooling began. The mechanical properties used in analysis are listed in Table 2 [38–40].

The warpage analysis during the PMC process was conducted after compression molding. The output results (deformation and stress) coming from the compression molding analysis were imported as an initial state in the PMC analysis. The viscoelastic properties over time (E (t)) were defined using data obtained from the stress relaxation test.

$$\mathrm{E(t)} = E_\infty + \sum_{i=1}^{N} E_i(t)e^{-t/\rho_i} \tag{8}$$

where E_∞, E_i, and ρ_i are the long-term modulus, Prony coefficient, and

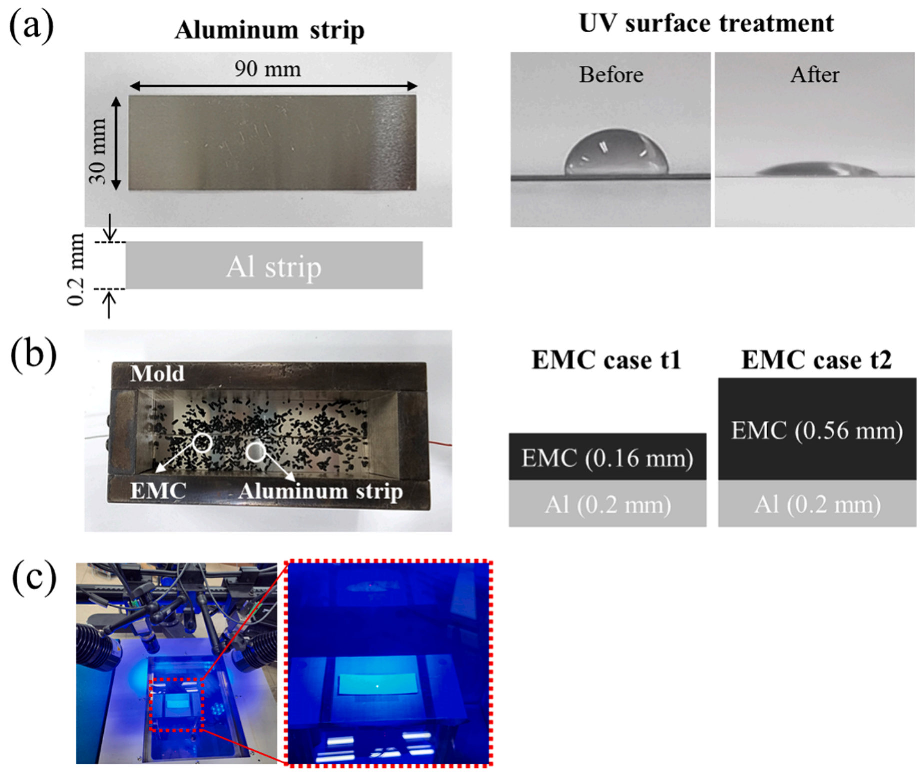

Fig. 8. (a) Aluminum strip geometry and water contact angle with respect to the UV surface treatment. (b) Fabrication process of Bi-material dumb package according to EMC thickness. (c) Warpage analysis using DIC during post-mold curing process.

relaxation time, respectively.

In the warpage analysis, the viscoelastic properties of the material were modeled using the Generalized Maxwell model, which provides a comprehensive representation of the time-dependent mechanical behavior. Both the shear (G(t)) and bulk modulus (K(t)) were expressed in the form of Prony series:

$$G(t) = G_\infty + \sum_{i=1}^{N} G_i(t)e^{-t/\rho_i} \tag{9}$$

$$K(t) = K_\infty + \sum_{i=1}^{N} K_i(t)e^{-t/\rho_i} \tag{10}$$

where, G and K are the long term shear modulus and bulk modulus. G_i and K_i are Prony coefficient. To determine the G(t) and K(t), we utilized the experimentally obtained relaxation modulus E(t) and assumed a constant Poisson's ratio (ν). The following relationships, derived from linear viscoelasticity theory, were employed: $G(t) = E(t)/2(1+\nu)$ and $K(t) = E(t)/3(1\text{-}2\nu)$.

Because the shift function exhibited different behavior below and above T_g, it was implemented through a UTRS. The shift function was defined as follows:

$$\log a = \frac{-C_1\left(T - T_{ref}\right)}{\left[C_2 + \left(T - T_{ref}\right)\right]} \tag{11}$$

where a, T, and T_{ref} are the shift factors, temperature, and reference temperature, respectively, and C_1 and C_2 are the constants of the Williams-Landel-Ferry (WLF) equation. It was also necessary to consider the cure shrinkage. Therefore, the expansion constant for thermal expansion and cure shrinkage was defined independently using a field variable (FV). The expansion constant was defined as the coefficient of thermal expansion (CTE) when the FV was 0 and as the cure shrinkage when the FV was 1. Because temperature-dependent deformation dominated during the heating and cooling processes, only the CTE was considered in this stage. However, thermal deformation due to temperature did not occur when the temperature was maintained. Therefore, in this case, only cure shrinkage was considered. In summary, during the heating and cooling process, FV was set to 0 to reflect the thermal shrinkage, and during the temperature hold, FV was set to 1 to reflect curing shrinkage. There was no external load in the PMC analysis, and only the temperature was varied according to the process profile. Gravity was assigned in the negative z direction for all analyses.

4.3. Warpage behavior of a bi-material dummy package

The warpage behavior of the bi-material strip based on the EMC thickness during the PMC process is shown in Fig. 10(a). Positive warpage (crying mode) was observed at room temperature where the PMC process began, and warpage deformation occurred in the negative direction (smile mode) as the temperature increased. This occurred because the CTE of aluminum was much larger than that of EMC, so the thermal strain generated in aluminum was much larger for the same temperature increase. It is noteworthy that a warpage change was observed even during the temperature dwelling period. Because there was no temperature change, this warpage behavior was not due to thermal deformation. Nevertheless, both the EMC t1 and t2 cases exhibited warpage deformation in the negative direction (smile mode), and the warpage deformation of the t2 case was about three times larger than that of t1 case.

There are two main factors that could induce internal strain changes during the temperature dwelling stage when deformation due to CTE mismatch does not occur: the viscoelastic behavior of EMC and EMC cure shrinkage. Regarding the viscoelastic properties of EMC, the absolute value of warpage should be reduced because the EMC internal stresses tend to relax over time. However, the experimental results showed that the amount of negative warpage deformation increased during the temperature dwelling period, indicating that the EMC cure

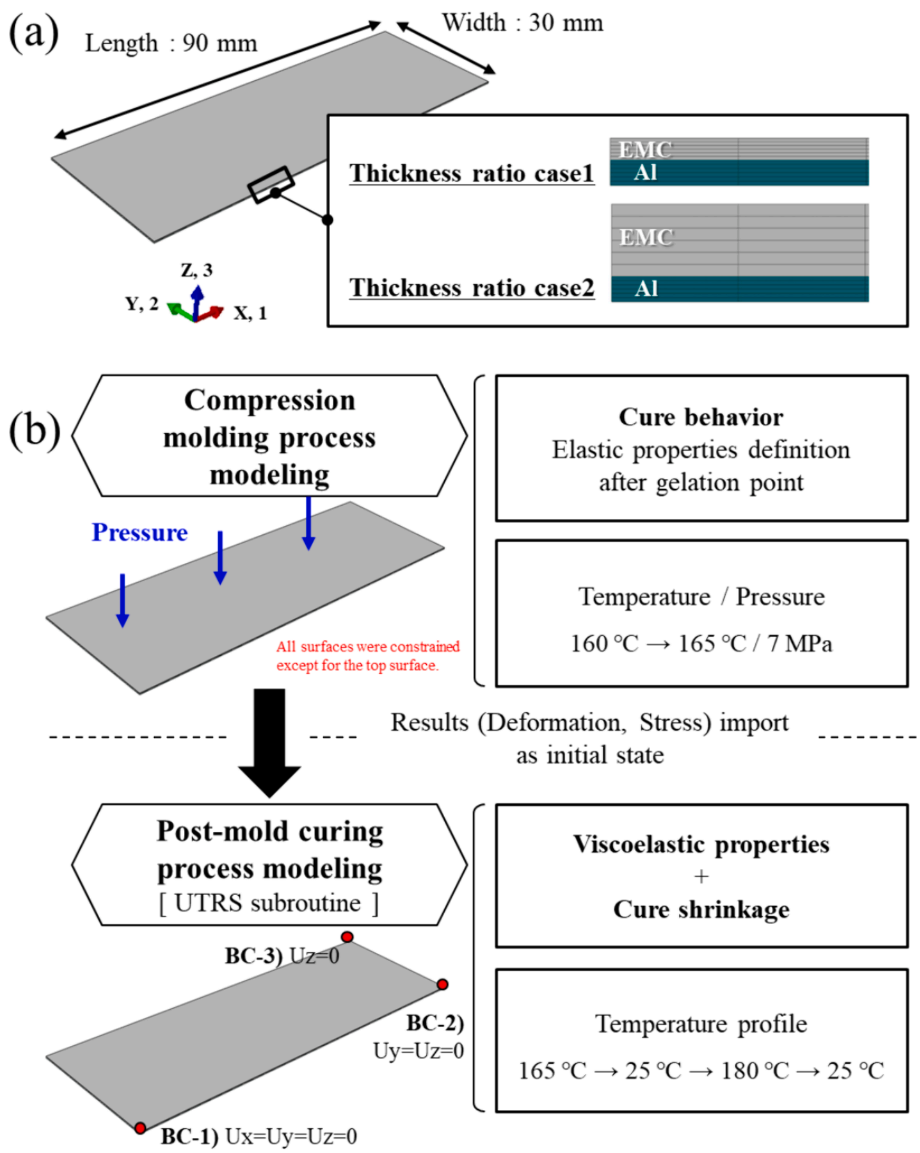

Fig. 9. (a) The bi-material dummy package model of WPG simulation according to the EMC thickness. (b) The flow chart of FEA process (Compression molding + Post-mold curing).

Table 2
Mechanical properties of EMC and aluminum for WPG simulation.

	Aluminum	EMC	
		Cure	PMC
Density [g/cm^3]	2.67	2.03	2.03
Poisson's ratio,υ	0.336	0.495–0.28	0.28
Elastic modulus [GPa]	68.08 @ 25 °C 62.08 @ 175 °C	0.06–2.143	Viscoelastic property
CTE, α_1 [ppm/°C]	23.1		9.4
CTE, α_2 [ppm/°C]	–		37.6
Glass transition temperature [°C]	–		145

shrinkage had a more significant influence than the stress relaxation effect. In both the compression molding and the PMC process, EMC cure shrinkage was an important factor affecting the warpage. Positive warpage deformation occurred during cooling, which is the last step of the PMC process, due to the difference of CTEs between the EMC and the aluminum.

As represented in Fig. 10(a), the FE analysis results (dotted line) showed quite similar behavior to the experimental results measured by DIC (solid line). To analyze the accuracy of warpage predictions more quantitatively, deformation shapes and warpage amounts were determined, as shown in Fig. 10(b) for the representative temperature points during the PMC process. Overall, the FE model successfully predicted both negative (smile mode) and positive (crying mode) warpage behaviors across all temperature ranges in the two specimens with

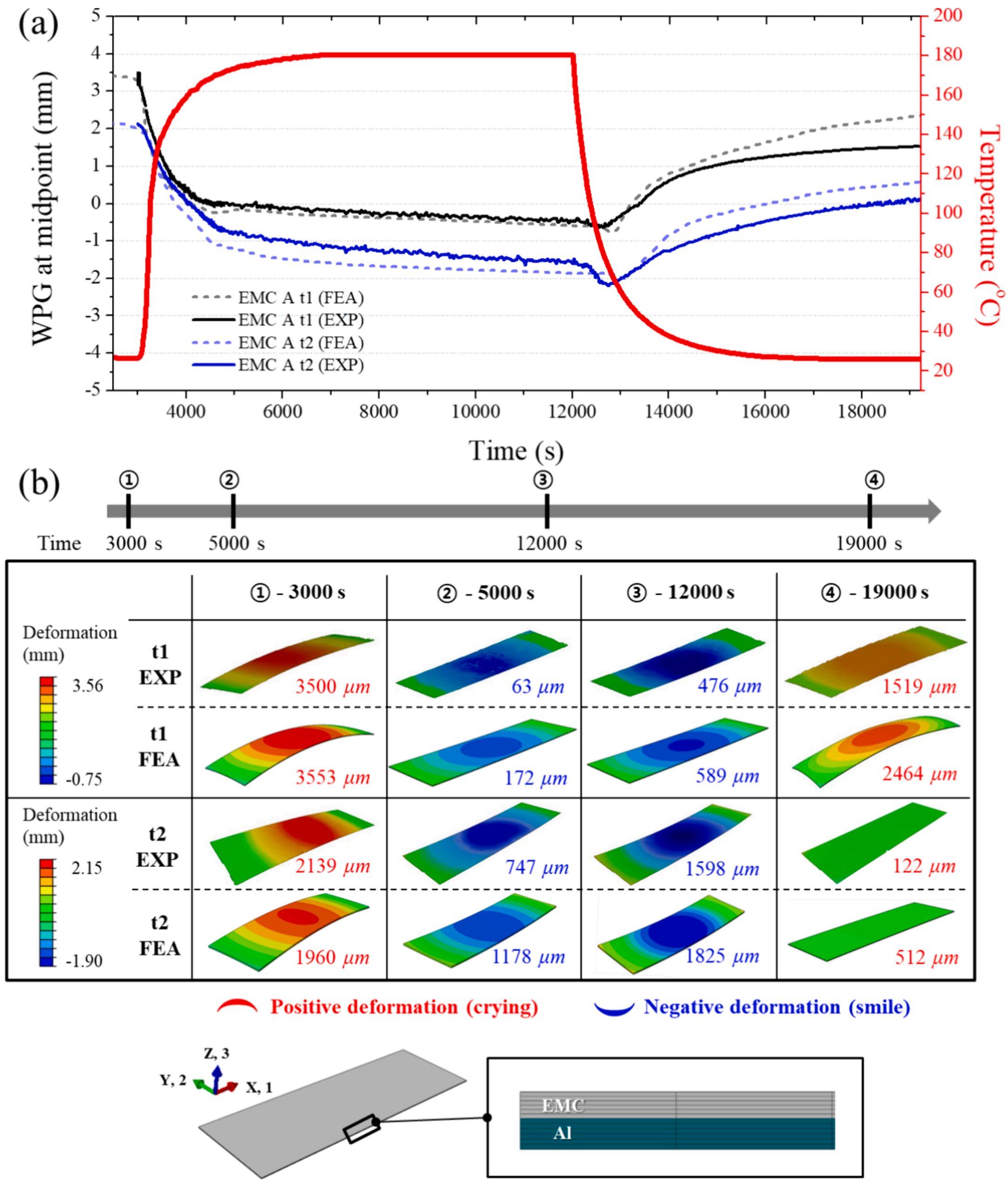

Fig. 10. WPG results of Bi-material dummy package during PMC process: (a) Temperature profile and WPG behavior according to experimental and finite element analysis (FEA) results. (b) Deformation contour obtained from DIC and FEA results.

different EMC thicknesses.

4.4. Temperature profile optimization for warpage reduction

To minimize the warpage occurring during the EMC curing process, optimization of the PMC temperature profile was performed, as indicated in the flowchart of Fig. 11(a). First, a warpage analysis model for the bi-material dummy package specimen was created using the FE analysis software ABAQUS. Subsequently, warpage simulations were conducted for eight arbitrary temperature profiles, including a linear temperature profile, and the final warpage amount was extracted from the output file. Following extraction of the final warpage amount, temperature profile optimization was performed using a genetic algorithm (GA), as indicated in Fig. 11(b). The GA initially ranked the final warpage values in ascending order, aiming for values closest to zero. Based on these rankings, a selection process was conducted to choose the top-performing individuals as parents. These selected data were then subjected to crossover operations for the temperature profile constants. Following the crossover, mutation was applied to the data (adding some random value), and a new generation was created by combining parents and offspring. This iterative process continued until warpage approached zero, which indicated optimization. Temperature profile optimization focused on the heating and cooling processes, excluding PMC temperature maintenance. To reduce the warpage using only the temperature profile without changing the process time duration, the temperature profile was determined at the fixed start and end points for each process, as shown in Fig. 11(c), because as the time increased, the warpage could decrease due to stress relaxation of the EMC. Additionally, one extra point was selected alongside the two fixed points, and the temperature profile was determined through interpolation using these three points for each process (heating and cooling). Thus, two points, representing the heating (t_H, T_H) and cooling (t_C, T_C) processes, were

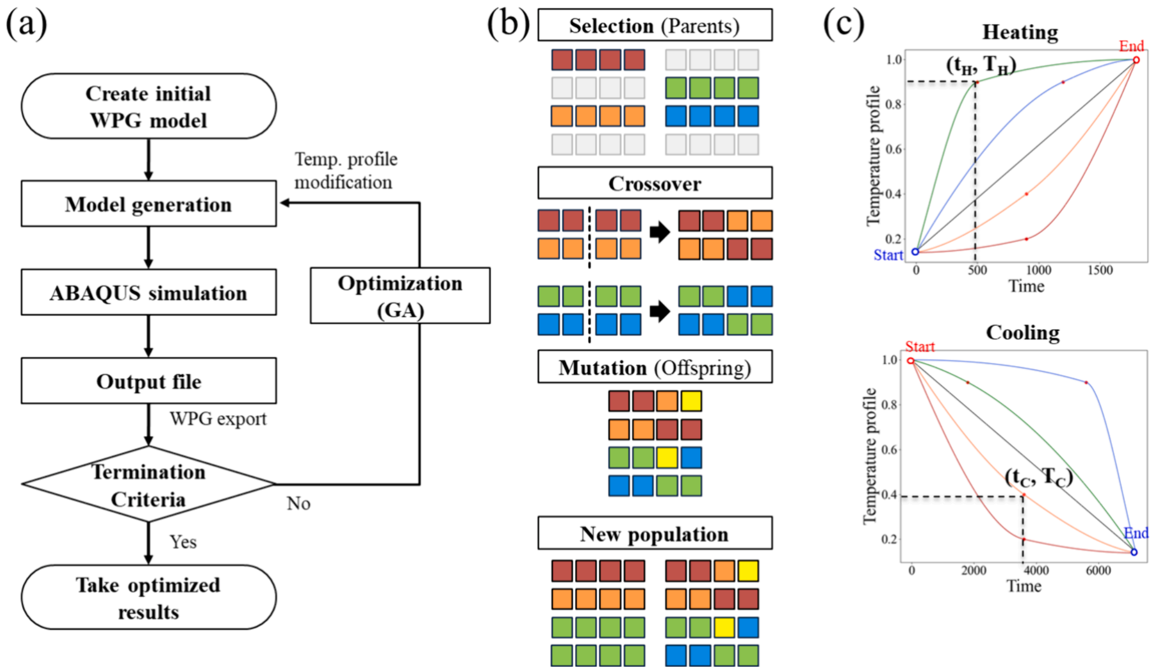

Fig. 11. (a) Flow chart of optimization process using GA. (b) Schematic of Genetic algorithm (GA) during the optimization process and each element represented the coefficient of temperature profile. (c) Heating and cooling temperature profiles according to the coefficients for the optimization. The y-axis means the normalized temperature value for the target temperature of PMC process.

used as temperature profile coefficients, and optimization was performed by changing the coefficients values. The optimization process was implemented using Python code with warpage simulation.

4.5. Optimization results for bi-material dummy packages

The warpage simulation of the bi-material dummy package specimens was performed according to the temperature profile, as shown in Fig. S3. Despite heating from the same initial temperature to the same target temperature over identical time intervals, very different warpage behaviors were observed. As the temperature increased, negative (smile) warpage deformation occurred, followed by a transition to positive (crying) warpage deformation at a specific point. The warpage direction change occurred near the EMC T_g. Furthermore, the temperature-dependent viscoelastic behavior of the epoxy matrix composite (EMC) led to variations in the deformation magnitude during the final stages of the heating step. These results confirmed that the temperature profile significantly affected the warpage behavior. Therefore, optimization of the PMC temperature profile was performed to minimize the amount of warpage utilizing these characteristics. The temperature profile of the heating and cooling stages varied according to the GA generation, starting from a linear temperature profile (see Fig. 12(a)). Also, the final amount of warpage decreased as the number of GA generations increased, as shown in Fig. 12(b). After optimizing for 100 generations, the final warpage amount was reduced by ~55 % compared to the initial state.

After conducting the optimization for 140 generations, the optimal temperature profile was obtained and was applied manually in the experiment, as shown in Fig. 13(a). During the PMC process, warpage was measured with respect to the temperature profile. In the optimized

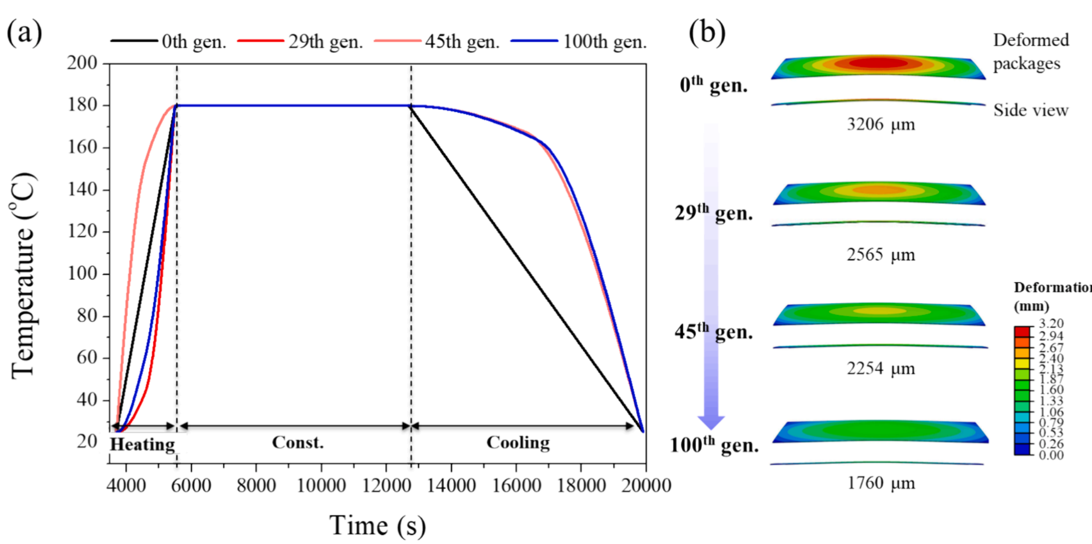

Fig. 12. (a) The change of temperature profile according to the GA generation. (b) Final warpage contour as a results of WPG simulation.

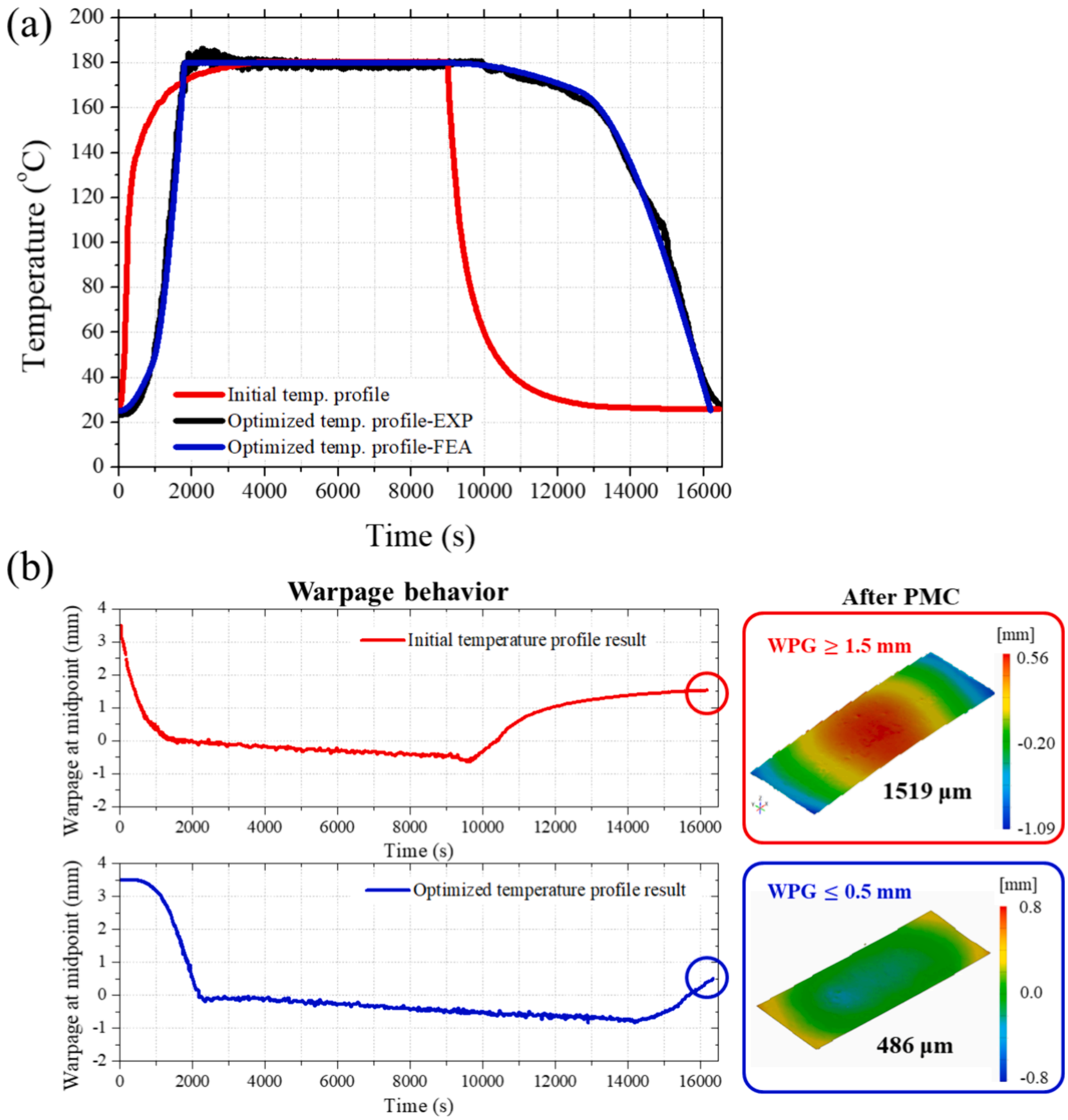

Fig. 13. (a) Temperature profile of PMC process before and after the optimization. (b) Warpage behavior of bi-material dummy package before and after the optimization.

temperature profile, warpage was gradually generated at the beginning of the heating stage and then increased abruptly, resulting in a different warpage value at the end of the heating stage (see Fig. 13(b)). After heating, almost zero warpage occurred in both cases (initial case and optimized case after 140 GA generations) and the warpage continuously increased in the negative direction (smile) until the end of the dwell time. During the cooling period, the warpage increased inversely in the positive direction (crying mode) in both cases. In the optimized temperature case, only 0.5 mm of final warpage at room temperature was generated, which was three times lower than the warpage amount of 1500 μm observed in the initial case with the linearly cooling profile case. In the initial temperature profile, the temperature dropped rapidly as cooling began, dropping below the T_g (145 °C) of EMC within 3 min. In contrast, in the optimized temperature profile, the temperature decreased gradually during the cooling stage, with a longer dwell time at high temperatures. Accordingly, this indicated that the stress inside the EMC could be sufficiently relieved by exposure to temperatures above the T_g for ~25 times longer than the initial temperature profile. Consequently, it was concluded that optimization of the temperature profile resulted in reduction of warpage to nearly zero, which indicated that adjustment in the process temperature profile could minimize warpage of the semiconductor package.

5. Conclusion

In this study, warpage in a bi-material strip (EMC/aluminum) dummy package specimen was successfully minimized by optimization of the temperature profile of the PMC process. A compression molding monitoring system was devised to analyze the shrinkage of the EMC during curing. Using this system, real-time temperature and strain monitoring was successfully performed during the compression molding process. Using a dielectric sensor, the gelation point of the EMC was determined to be ~160 °C. As a result, cure shrinkage began to occur after the gelation point in the compression molding process, and a large strain of 0.39 % occurred during the curing process. When pressure was removed under process temperature conditions, the strain decreased to 0.30 % due to recovery of elastic strain. Furthermore, cure shrinkage during the PMC process was measured using FBG sensors, resulting in a strain of ~0.05 %, lower than that during compression molding. Temperature profile optimization using GA was performed based on the developed warpage simulation model. When the bi-material strip (EMC/aluminum) dummy package specimen was fabricated using the optimal temperature profile, a three-fold reduction in warpage was achieved compared to the previous temperature profile. Consequently, it is expected that the temperature profile optimization technique proposed in this work can be used for reduction of the warpage of real package structures in the semiconductor industry.

CRediT authorship contribution statement

Hui-Jin Um: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Young-Min Ju:** Methodology, Investigation, Data curation. **Dae-Woong Lee:** Supervision, Resources. **Jinho Ahn:** Supervision. **Hak-Sung Kim:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This research was supported by a Semiconductor Industry Collaborative Project between Hanyang University and SK hynix Inc. This work was also supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. RS-2023-00260527). This work was also partly supported by Korea Evaluation Institute of Industrial Technology (KEIT) grant funded by the Korea government (MOTIE) (Project No. 20014863, Development of 3D formable heating elements and forming technology for energy-efficient radiant heating of electric vehicles).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matdes.2024.113265.

REFERENCES

[1] M. van Dijk, S. Huber, A. Stegmaier, H. Walter, O. Wittler, M. Schneider-Ramelow, Experimental and simulative study of warpage behavior for fan-out wafer-level packaging, Microelectron. Reliab. 135 (2022) 114585.
[2] G. Yang, Z. Kuang, H. Lai, Y. Liu, R. Cui, J. Cao, Y. Zhang, C. Cui, A quantitative model to understand the effect of gravity on the warpage of fan-out panel-level packaging, IEEE Trans. Compon. Packag. Manuf. Technol. 11 (11) (2021) 2022–2030.
[3] G. Hu, J.-E. Luan, S. Chew, Characterization of chemical cure shrinkage of epoxy molding compound with application to warpage analysis, (2009).
[4] C. Chen, D. Yu, T. Wang, Z. Xiao, L. Wan, Warpage prediction and optimization for embedded silicon fan-out wafer-level packaging based on an extended theoretical model, IEEE Trans. Compon. Packag. Manuf. Technol. 9 (5) (2019) 845–853.
[5] M.-L. Wu, J.-S. Lan, Simulation and experimental study of the warpage of fan-out wafer-level packaging: The effect of the manufacturing process and optimal design, IEEE Trans. Compon. Packag. Manuf. Technol. 9 (7) (2018) 1396–1405.
[6] C. Xu, P. Sun, Z. Liu, J. Liu, Wafer Compression Molding Warpage Optimization for Wafer Level Package, 2020 21st International Conference on Electronic Packaging Technology (ICEPT), IEEE, 2020, pp. 1-4.
[7] F. Qin, S. Zhao, Y. Dai, M. Yang, M. Xiang, D. Yu, Study of warpage evolution and control for six-side molded WLCSP in different packaging processes, IEEE Trans. Compon. Packag. Manuf. Technol. 10 (4) (2020) 730–738.
[8] N. Garg, Y.H. Leong, D. Tan, T. Tsuji, Study of raw materials effect on the dielectric properties of epoxy molding compound for high voltage power devices, 2019 IEEE 21st electronics packaging technology conference (EPTC), IEEE 2019 (2019) 305–310.
[9] J.-W. Bae, W. Kim, S. Hwang, Y.-S. Choe, S.-H. Lee, A low-viscousity, highly thermally conductive epoxy molding compound (EMC), Macromol. Res. 12 (2004) 78–84.
[10] Y. Wang, B. Han, D. Kim, A. Bar-Cohen, P. Joseph, Integrated measurement technique for curing process-dependent mechanical properties of polymeric materials using fiber bragg grating, Exp. Mech. 48 (2008) 107–117.
[11] Y. Wang, L. Woodworth, B. Han, Simultaneous measurement of effective chemical shrinkage and modulus evolutions during polymerization, Exp. Mech. 51 (2011) 1155–1169.
[12] T.-C. Chiu, H.-W. Huang, Y.-S. Lai, Effect of chemical aging on warpage for encapsulated packages—characterization and simulation, 2011 IEEE 61st electronic components and technology conference (ECTC), IEEE 2011 (2011) 1894–1900.
[13] M.-Y. Lin, Y.-J. Zeng, S.-J. Hwang, M.-H. Wang, H.-P. Liu, C.-L. Fang, Warpage and residual stress analyses of post-mold cure process of IC packages, Int. J. Adv. Manuf. Technol. 124 (3) (2023) 1017–1039.
[14] M. Sadeghinia, K.M. Jansen, L.J. Ernst, Characterization of the viscoelastic properties of an epoxy molding compound during cure, Microelectron. Reliab. 52 (8) (2012) 1711–1718.
[15] M. Sadeghinia, K. Jansen, L. Ernst, Characterization and modeling the thermo-mechanical cure-dependent properties of epoxy molding compound, Int. J. Adhes. Adhes. 32 (2012) 82–88.
[16] P. Gromala, B. Muthuraman, B. Öztürk, K. Jansen, L. Ernst, Material characterization and nonlinear viscoelastic modelling of epoxy based thermosets for automotive application, in: 2015 16th International Conference on Thermal, Mechanical and Multi-Physics Simulation and Experiments in Microelectronics and Microsystems, IEEE, 2015, pp. 1–7.
[17] M.R. Sangtabi, M.S. Kiasat, Long-term viscoelastic properties of an adhesive and molding compound, characterization and modeling, Polymer 116 (2017) 204–217.
[18] H.-W. Huang, T.-C. Chiu, Y.-S. Lai, Warpage simulation for encapsulated package during post-mold curing process, 2010 5th International Microsystems Packaging Assembly and Circuits Technology Conference, IEEE 2010 (2010) 1–4.
[19] T. Hasegawa, H. Abe, T. Ikeuchi, Wafer level compression molding compounds, 2012 IEEE 62nd Electronic Components and Technology Conference, IEEE, 2012, pp. 1400-1405.
[20] H.-C. Cheng, Y.-C. Liu, Warpage characterization of molded wafer for fan-out wafer-level packaging, J. Electron. Packag. 142 (1) (2020) 011004.
[21] N. Lorenz, M. Müller-Pabel, J. Gerritzen, J. Müller, B. Gröger, D. Schneider, K. Fischer, M. Gude, C. Hopmann, Characterization and modeling cure-and pressure-dependent thermo-mechanical and shrinkage behavior of fast curing epoxy resins, Polym. Test. 108 (2022) 107498.
[22] C. Tsai, S. Liu, K. Chiang, Warpage analysis of fan-out panel-level packaging using equivalent CTE, IEEE Trans. Device Mater. Reliab. 20 (1) (2019) 51–57.
[23] C.-Y. Huang, D. Ng, H.-H. Lee, V. Lin, C.-F. Lin, C.K. Chung, Process induced wafer warpage optimization for multi-chip integration on wafer level molded wafer, 2019 IEEE 69th Electronic Components and Technology Conference (ECTC), IEEE, 2019, pp. 1287-1293.
[24] J. Chao, R. Zhang, T. Do, A. Tong, Y. Ma, D. Grimes, R.K. Trichur, L. Bao, Low Warpage Liquid Compression Molding (LCM) Material for High Density Fan-out and Wafer Level Packaging Applications, 2020 IEEE 70th Electronic Components and Technology Conference (ECTC), IEEE, 2020, pp. 924-930.
[25] F.X. Che, D. Ho, T.C. Chai, Study on warpage and reliability of fan-out interposer technology, IEEE Trans. Compon. Packag. Manuf. Technol. 9 (4) (2018) 786–796.
[26] P. Sun, V.C.-K. Leung, B. Xie, V.W. Ma, D.X.-Q. Shi, Warpage reduction of package-on-package (PoP) module by material selection & process optimization, 2008 International Conference on Electronic Packaging Technology & High Density Packaging, ieee, 2008, pp. 1-6.
[27] S.P. Phansalkar, C. Kim, B. Han, Effect of critical properties of epoxy molding compound on warpage prediction: A critical review, Microelectron. Reliab. 130 (2022) 114480.
[28] H.S. Kim, Reduction of fabricational thermal residual stress of the hybrid co-cured structure using a dielectrometry, Compos. Sci. Technol. 67 (1) (2007) 29–44.
[29] J.-H. Baek, D.-W. Park, G.-H. Oh, D.-O. Kawk, S.S. Park, H.-S. Kim, Effect of cure shrinkage of epoxy molding compound on warpage behavior of semiconductor package, Mater. Sci. Semicond. Process. 148 (2022) 106758.
[30] D. Zhang, J. Wang, Y. Wang, X. Dai, A fast response temperature sensor based on fiber Bragg grating, Meas. Sci. Technol. 25 (7) (2014) 075105.

[31] R. Li, Y. Tan, Y. Chen, L. Hong, Z. Zhou, Investigation of sensitivity enhancing and temperature compensation for fiber Bragg grating (FBG)-based strain sensor, Opt. Fiber Technol. 48 (2019) 199–206.
[32] G. Yan, T. Wang, L. Zhu, F. Meng, W. Zhuang, A novel strain-decoupled sensitized FBG temperature sensor and its applications to aircraft thermal management, Opt. Laser Technol. 140 (2021) 106597.
[33] B. Zhang, M. Kahrizi, High-temperature resistance fiber Bragg grating temperature sensor fabrication, IEEE Sens. J. 7 (4) (2007) 586–591.
[34] J. Kumar, G. Singh, M.K. Saxena, S.K. Dixit, S.V. Nakhe, Development and studies on FBG temperature sensor for applications in nuclear fuel cycle facilities, IEEE Sens. J. 21 (6) (2020) 7613–7619.
[35] S. ASTM, Standard test methods for flexural properties of unreinforced and reinforced plastics and electrical insulating materials. ASTM D790, Annual book of ASTM Standards (1997).
[36] J.S. Kim, D.G. Lee, Measurement of the degree of cure of carbon fiber epoxy composite materials, J. Compos. Mater. 30 (13) (1996) 1436–1457.
[37] S. Timoshenko, Analysis of bi-metal thermostats, Josa 11 (3) (1925) 233–255.
[38] S. Minakuchi, Y. Okabe, N. Takeda, Real-time detection of debonding between honeycomb core and facesheet using a small-diameter FBG sensor embedded in adhesive layer, J. Sandw. Struct. Mater. 9 (1) (2007) 9–33.
[39] C. Wen, Y. Li, Effects of metal coating on the fiber Bragg grating temperature sensing characteristics, J. Mod. Opt. 63 (8) (2016) 762–770.
[40] W. Shen, X. Wang, L. Xu, Y. Zhao, Strain transferring mechanism analysis of the substrate-bonded FBG sensor, Optik 154 (2018) 441–452.